

WEEK 3 PROJECT REPORT

Financial Data Import, Transformation & Time Intelligence

Dataset: Shopify Stock Data

1. Introduction

This project focuses on analyzing Shopify stock data using Microsoft Power BI. The dataset contains historical stock information including date, opening price, high price, low price, closing price, and trading volume.

Power BI was used to import and prepare the data, create a dedicated Date Table, establish a relationship between the tables, create DAX measures, build a custom date hierarchy, perform time-intelligence calculations, and create an interactive dashboard.

2. Objectives

The main objectives of this project were:

- To import and prepare Shopify stock data in Power BI.
- To create a dedicated Date Table using DAX.
- To establish a relationship between the Date Table and the stock dataset.
- To create financial and time-intelligence measures using DAX.
- To create a custom Year-Quarter-Month-Day hierarchy.
- To create visualizations and an interactive dashboard for Shopify stock analysis.

3. Dataset

The dataset used for this project is Shopify Stock Data.

The main fields used in the dataset are:

- Date
- Open
- High
- Low
- Close
- Volume

These fields were used for creating the calculations and visualizations in Power BI.

Dataset Import

Shop Dataset.csv

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

date	open	high	low	close	adj_close	volume
21-05-2015 00:00:00 -04:00	2.799999952	2.874000072	2.411000013	2.568000078	2.568000078	123039000
22-05-2015 00:00:00 -04:00	2.607000113	3.109999895	2.599999905	2.831000009	2.831000009	28412000
26-05-2015 00:00:00 -04:00	2.980000019	3.03399992	2.907999992	2.964999914	2.964999914	8202000
27-05-2015 00:00:00 -04:00	3.066999912	3.08100009	2.700000048	2.75	2.75	7976000
28-05-2015 00:00:00 -04:00	2.755000114	2.773999929	2.648000002	2.744999886	2.744999886	4000000
29-05-2015 00:00:00 -04:00	2.724999905	2.737999916	2.687000036	2.720000029	2.720000029	2554000
01-06-2015 00:00:00 -04:00	2.720000029	2.743999958	2.660000086	2.733000004	2.733000004	2569000
02-06-2015 00:00:00 -04:00	2.746999979	2.746999979	2.664999962	2.694999933	2.694999933	3798000
03-06-2015 00:00:00 -04:00	2.676000118	2.75	2.650000095	2.743999958	2.743999958	3024000
04-06-2015 00:00:00 -04:00	2.742000103	2.749000072	2.551000118	2.677999973	2.677999973	3226000
05-06-2015 00:00:00 -04:00	2.665999889	2.739000082	2.609999895	2.739000082	2.739000082	1294000
08-06-2015 00:00:00 -04:00	2.690000057	2.747999907	2.681999922	2.697999954	2.697999954	2809000
09-06-2015 00:00:00 -04:00	2.664000034	2.845999956	2.664000034	2.834000111	2.834000111	3375000
10-06-2015 00:00:00 -04:00	2.845000029	2.888000011	2.795000076	2.809999943	2.809999943	2734000
11-06-2015 00:00:00 -04:00	2.848999977	3.072999954	2.848999977	3.072999954	3.072999954	4606000
12-06-2015 00:00:00 -04:00	3.131999969	3.369999886	3.125	3.362999916	3.362999916	8474000
15-06-2015 00:00:00 -04:00	3.397000074	3.812000036	3.292000055	3.549999952	3.549999952	16943000
16-06-2015 00:00:00 -04:00	3.605999947	3.837899923	3.584000111	3.811000109	3.811000109	11095000
17-06-2015 00:00:00 -04:00	3.963000059	4.212999821	3.801000118	3.862999916	3.862999916	15952000
18-06-2015 00:00:00 -04:00	3.842000008	3.888999939	3.432600021	3.555000067	3.555000067	11482000

The data in the preview has been truncated due to size limits.

Extract Table Using Examples

Load

Transform Data

Cancel

4. Data Import and Transformation

The Shopify stock dataset was imported into Power BI for analysis.

The data was examined and prepared for use in the Power BI model. The date field was also prepared for creating the relationship with the dedicated Date Table.

Power Query Editor was used during the data preparation process.

Data Transformation

date	1.2 high	1.2 low	1.2 close	1.2 adj_close	1.2 volume
21-05-20	2.874000072	2.411000013	2.568000078	2.568000078	12303
22-05-20	3.109999895	2.599999905	2.83100009	2.83100009	2841
26-05-20	3.03399992	2.907999992	2.964999914	2.964999914	820
27-05-20	3.08100009	2.700000048	2.75	2.75	797
28-05-20	2.773999929	2.648000002	2.744999886	2.744999886	400
29-05-20	2.737999916	2.687000036	2.720000029	2.720000029	255
01-06-20	2.743999958	2.660000086	2.733000004	2.733000004	256
02-06-20	2.746999979	2.664999962	2.694999933	2.694999933	379
03-06-20	2.75	2.650000095	2.743999958	2.743999958	302
04-06-20	2.749000072	2.551000118	2.677999973	2.677999973	322
05-06-20	2.739000082	2.609999895	2.739000082	2.739000082	129
08-06-20	2.747999907	2.681999922	2.697999954	2.697999954	280
09-06-20	2.845999956	2.664000034	2.834000111	2.834000111	337
10-06-20	2.888000011	2.795000076	2.809999943	2.809999943	273
11-06-20	3.072999954	2.848999977	3.072999954	3.072999954	460
12-06-20	3.369999886	3.125	3.362999916	3.362999916	847
15-06-20	3.812000036	3.292000055	3.549999952	3.549999952	1694
16-06-20	3.837899923	3.584000111	3.811000109	3.811000109	1109
17-06-20	4.212999821	3.801000118	3.862999916	3.862999916	1595
18-06-20	3.888999939	3.432600021	3.555000067	3.555000067	1148
19-06-2015 00:00:00 -04:00	3.632999897	3.674000025	3.255000114	3.552000046	979
22-06-2015 00:00:00 -04:00	3.676000118	3.849999905	3.608999968	3.694000006	792
23-06-2015 00:00:00 -04:00	3.775000095	3.809999943	3.516999996	3.542999983	293
24-06-2015 00:00:00 -04:00	3.512000084	3.742899895	3.414000034	3.595999956	564
25-06-2015 00:00:00 -04:00	3.634000063	3.650000095	3.480000019	3.483999968	208
26-06-2015 00:00:00 -04:00	3.485999989	3.532000065	3.263000011	3.351999998	346

5. Creation of Date Table

A dedicated Date Table was created in Power BI using the CALENDAR() function.

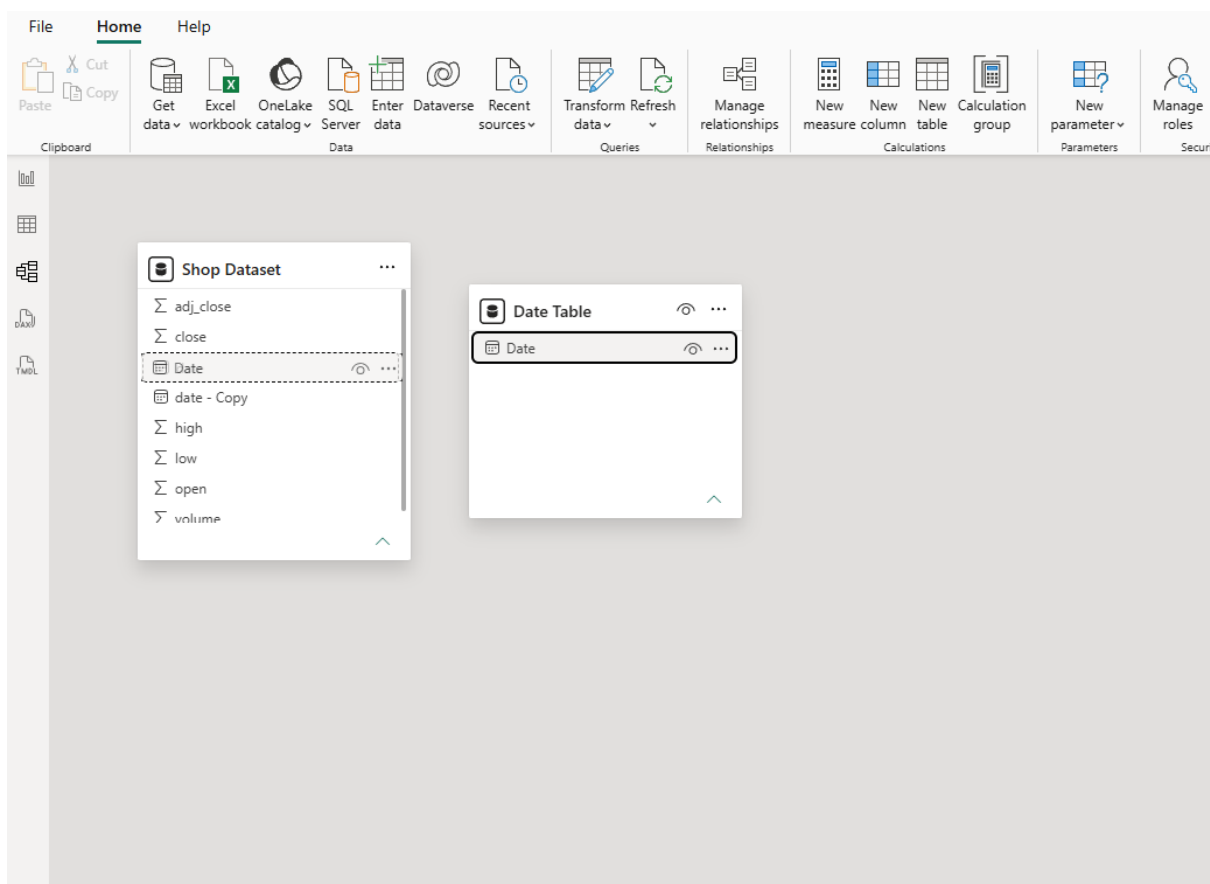
The Date Table contains the following fields:

- Date
- Day

- Month
- Month Number
- Quarter
- Year

The Date Table was created to support date-based analysis and time-intelligence calculations.

Creation of Date Table



6. Marking the Date Table as the Official Date Table

The created Date Table was marked as the official Date Table in Power BI.

The Date column was validated successfully for use as the date column.

Mark as Date Table

Mark as a date table ✕

To enable the creation of date-related visuals, tables and quick measures using this table's date data, mark it as a date table.

Keep in mind any built-in date tables that are already associated with this table will be removed. Visuals or DAX expressions referring to them may break. [Learn more](#)

Mark as a date table

☒ On

Choose a date column

Date

✔ Validated successfully

Save

Cancel

7. Relationship Between Date Table and Shop Dataset

A relationship was created between the Date Table and Shop Dataset.

The relationship settings used were:

- From Table: Date Table
- Column: Date
- To Table: Shop Dataset
- Column: date
- Cardinality: Many to One (*:1)

- Cross-filter Direction: Single
- Active Relationship: Yes

The Date Table is on the one side, while the Shop Dataset is on the many side.

Relationship

Edit relationship



Select tables and columns that are related.

From table

Date Table

Date	Day	Month	Month Number	Quarter	Year
21-05-2015 0...	21	May	5	Q2	2015
22-05-2015 0...	22	May	5	Q2	2015
23-05-2015 0...	23	May	5	Q2	2015

To table

Shop Dataset

adj_close	close	date	date - Copy	high	low	open
2.5680000782...	2.5680000782...	21-05-2015 0...	21 May 2015	2.8740000724...	2.4110000133...	2.79999...
2.8310000896...	2.8310000896...	22-05-2015 0...	22 May 2015	3.1099998950...	2.5999999046...	2.60700...
2.9649999141...	2.9649999141...	26-05-2015 0...	26 May 2015	3.0339999198...	2.9079999923...	2.98000...

Cardinality

Many to one (*:1)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

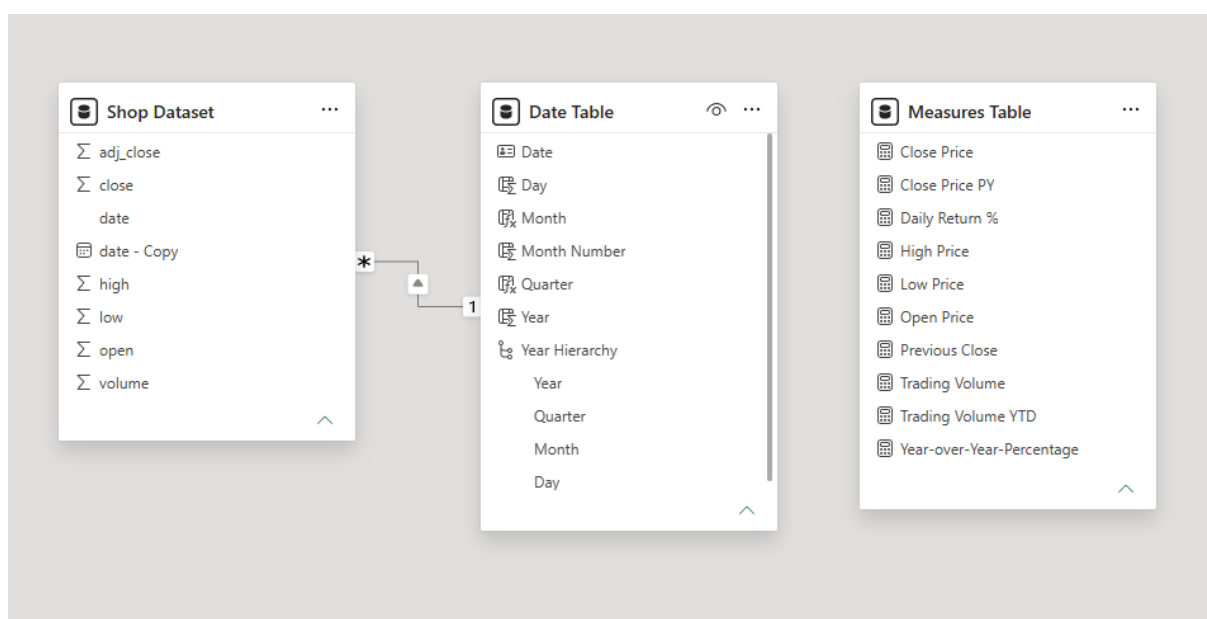
Cancel

8. Data Model

The completed Power BI model contains the Date Table, Shop Dataset, and Measures Table.

The relationship between the Date Table and Shop Dataset allows the stock data to be analyzed using the date fields from the Date Table.

Relationship



9. Custom Date Hierarchy

A custom date hierarchy named Year Hierarchy was created.

The hierarchy contains:

Year → Quarter → Month → Day

This hierarchy was created to support drill-down analysis across different levels of time.

10. DAX Measures

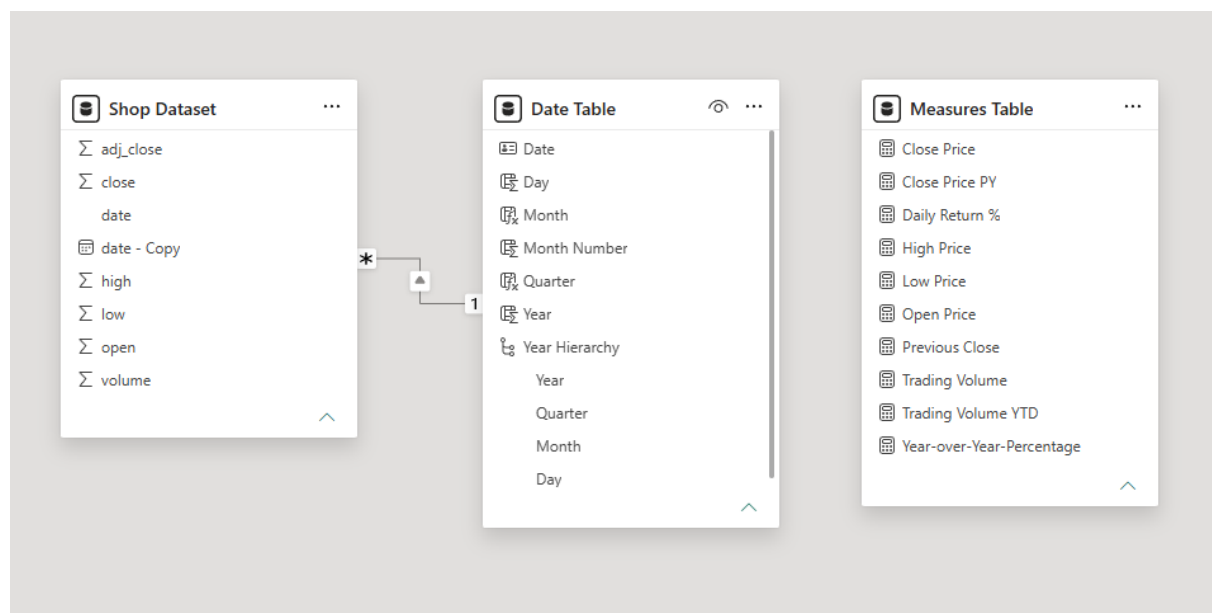
DAX measures were created for analyzing the Shopify stock data.

The Measures Table contains the following measures:

1. High Price
2. Low Price
3. Close Price
4. Close Price PY
5. Daily Return %
6. Open Price
7. Previous Close
8. Trading Volume
9. Trading Volume YTD
10. Year-over-Year-Percentage

These measures were used in the cards, charts, and time-intelligence analysis in the dashboard.

Measures Table



11. Previous Close Calculation

The Previous Close measure was created using the DATEADD() function to retrieve the closing price from the previous day.

The measure used was:

Previous Close =

```
CALCULATE(  
    MAX('Shop Data'[close]),  
    DATEADD('Date Table'[Date], -1, DAY)  
)
```

12. Daily Return Percentage

The Daily Return % measure was created by comparing the current closing price with the previous closing price.

The measure used was:

Daily Return % =

```
DIVIDE(  
    MAX('Shop Data'[close]) - [Previous Close],  
    [Previous Close],  
    0  
)
```

13. Time-Intelligence Measures

Time-intelligence calculations were created as part of the Week 3 analysis.

The measures created include:

- Close Price PY
- Previous Close
- Trading Volume YTD
- Year-over-Year-Percentage

These measures were used to compare stock values across different dates and periods.

14. Dashboard Visualizations

The created measures and fields were used to build the following Power BI visualizations:

Trading Volume Over Time

A stacked column chart was created to display trading volume over time.

X-axis: Date

Y-axis: Trading Volume

Close Price Trend Over Time

A line chart was created to display the closing price trend over time.

X-axis: Date

Y-axis: Close Price

High Price vs Low Price

A line chart was created to compare the high and low stock prices over time.

X-axis: Date

Y-axis: High Price and Low Price

Close Price vs Previous Year

A line chart was created to compare the current Close Price with the Close Price from the previous year.

X-axis: Date

Y-axis: Close Price and Close Price PY

15. KPI Cards

Five KPI cards were created for the dashboard:

- Close Price
- High Price
- Low Price
- Daily Return %
- Trading Volume

These cards provide a quick view of the selected stock indicators.

16. Year Slicer

A Year slicer was added to the dashboard using the Year field from the Date Table.

The slicer was changed to a Dropdown format so that a year can be selected from the list.

The slicer allows the dashboard data to be filtered by year.

17. Final Dashboard

The final Power BI dashboard combines the created charts, KPI cards, and Year slicer into one report page.

The dashboard contains:

- Trading Volume Over Time
- Close Price Trend Over Time
- Close Price Card
- High Price Card
- Low Price Card
- Daily Return % Card
- Trading Volume Card
- High Price vs Low Price
- Close Price vs Previous Year
- Year Slicer

Final Dashboard



18. Conclusion

The Shopify stock dataset was successfully imported and prepared in Power BI. A dedicated Date Table was created and connected to the Shop Dataset using an active many-to-one relationship.

DAX measures were created for stock prices, trading volume, daily return, previous close, previous-year comparison, YTD analysis, and year-over-year analysis. A Year-Quarter-Month-Day hierarchy was also created for time-based analysis.

Finally, the measures and fields were used to create KPI cards, charts, and a Year slicer, resulting in an interactive Shopify Stock Analysis dashboard.